

CS 6341 Robotics: Ubuntu Terminal Basics for ROS 2

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September 3, 2026

ROS 2 development requires frequent use of the Ubuntu terminal. This tutorial introduces some basic Linux commands that you will use throughout the course.

You do not need to become a Linux expert. However, you should be comfortable navigating directories, creating and editing files, installing packages, and running ROS 2 commands from the terminal.

1 Opening a Terminal

On Ubuntu, you can open a terminal from the application menu.

A useful keyboard shortcut is:

`Ctrl+Alt+T`

If you are using the Docker environment for this course, open a terminal inside the Ubuntu desktop displayed in your web browser.

You will see a command prompt similar to:

```
ubuntu@computer:~$
```

You type commands after the \$ symbol. Do not type the \$ itself.

2 Understanding the File System

Unlike Windows, which commonly uses paths such as

```
C:\data\ros2_ws
```

Ubuntu uses paths such as

```
/home/ubuntu/data/ros2_ws
```

The forward slash

```
/
```

represents the root of the Linux file system.

2.1 Your Home Directory

The symbol

~

is a shortcut for your home directory.

For example:

~/data

means the data directory inside your home directory.

You can display your home directory with:

echo \$HOME

3 Finding Your Current Directory

Use:

pwd

The command stands for **print working directory**.

For example, it may display:

/home/ubuntu/data/ros2_ws

Whenever you are unsure where you are in the file system, use pwd.

4 Listing Files and Directories

Use:

ls

to display files and directories in your current location.

For more information, use:

ls -l

To also display hidden files:

ls -a

You can combine options:

ls -la

5 Changing Directories

Use `cd` to change directories.

For example:

```
cd ~/data
```

To enter your ROS 2 workspace:

```
cd ~/data/ros2_ws
```

To enter the `src` directory:

```
cd src
```

To move up one directory:

```
cd ..
```

To return directly to your home directory:

```
cd ~
```

or simply:

```
cd
```

6 Absolute and Relative Paths

An **absolute path** starts from the root of the file system:

```
/home/ubuntu/data/ros2_ws/src
```

A **relative path** is interpreted relative to your current directory.

For example, if you are currently in:

```
/home/ubuntu/data/ros2_ws
```

then:

```
cd src
```

is equivalent to:

```
cd /home/ubuntu/data/ros2_ws/src
```

The symbol

```
.
```

means the current directory, while:

```
..
```

means the parent directory.

7 Creating Directories

Use `mkdir` to create a directory.

For example:

```
mkdir data
```

To create multiple levels of directories at once, use `-p`:

```
mkdir -p ~/data/ros2_ws/src
```

This command creates any missing directories in the path.

8 Creating and Viewing Files

You can create an empty file using:

```
touch test.txt
```

To display the contents of a text file:

```
cat test.txt
```

For longer files, use:

```
less test.txt
```

Press `q` to exit `less`.

9 Copying, Moving, and Renaming Files

To copy a file:

```
cp file1.txt file2.txt
```

To copy a directory and everything inside it:

```
cp -r folder1 folder2
```

To move a file:

```
mv file.txt ~/data/
```

The `mv` command is also used to rename files:

```
mv old_name.txt new_name.txt
```

10 Deleting Files and Directories

To delete a file:

```
rm file.txt
```

To delete a directory and everything inside it:

```
rm -r directory_name
```

Be careful with rm. Files removed from the terminal are generally not moved to the Trash.

In particular, be very careful when using:

```
rm -rf directory_name
```

This recursively deletes files without asking for confirmation.

11 Tab Completion

One of the most useful terminal features is **Tab completion**.

Suppose you have a directory named:

```
~/data/ros2_ws
```

Instead of typing the entire path, you can type:

```
cd ~/data/ros
```

and press the Tab key.

The terminal may automatically complete it to:

```
cd ~/data/ros2_ws/
```

Use Tab completion frequently. It saves time and helps avoid typing errors.

12 Command History

Press the **Up Arrow** key to see commands that you previously executed.

Press the **Down Arrow** key to move forward through the command history.

You can also display your command history with:

```
history
```

To search previous commands interactively, press:

```
Ctrl+R
```

and begin typing part of a previous command.

13 Stopping a Running Program

Many terminal programs continue running until you stop them.

For example:

```
rviz2
```

or a ROS 2 node may continue running in the terminal.

To stop the current program, press:

Ctrl+C

You will use Ctrl+C frequently when working with ROS 2.

14 Installing Ubuntu Packages

Ubuntu uses the apt package manager to install software.

First, update the package information:

```
sudo apt update
```

Then install a package. For example:

```
sudo apt install ros-jazzy-urdf-launch
```

The sudo command runs a command with administrator privileges.

You may be asked for your password when using sudo.

14.1 When Should You Use sudo?

Use sudo when installing or modifying system software, for example:

```
sudo apt install <package>
```

Do **not** automatically add sudo to every command.

For example, when building your ROS 2 workspace, use:

```
colcon build
```

Do **not** use:

```
sudo colcon build
```

15 Git Basics

We will use Git to download source code from GitHub.

To clone a repository:

```
git clone <repository_url>
```

For example:

```
cd ~/data/ros2_ws/src  
git clone -b ros2 https://github.com/ros/urdf_tutorial.git
```

After cloning, you can verify that the directory exists:

```
ls
```

You should see:

```
urdf_tutorial
```

16 Environment Variables

ROS 2 uses environment variables to locate installed packages and configure the ROS environment.

For example:

```
echo $ROS_DISTRO
```

displays the current ROS distribution.

After sourcing ROS 2 Jazzy:

```
source /opt/ros/jazzy/setup.bash
```

the following command:

```
echo $ROS_DISTRO
```

should display:

```
jazzy
```

17 What Does source Mean?

You will frequently see the command:

```
source /opt/ros/jazzy/setup.bash
```

The source command executes a script in your current shell.

For ROS 2, this configures your terminal so that it knows where ROS 2 commands and packages are located.

After building your own workspace, you will also run:

```
source ~/data/ros2_ws/install/setup.bash
```

This makes the packages in your workspace available to ROS 2.

If you open a new terminal, you need to source the appropriate setup files again unless you have added them to your .bashrc.

18 The .bashrc File

The file

```
~/ .bashrc
```

contains commands that are automatically executed when you open a new Bash terminal.

For example, you can add:

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

After your workspace has been built, you can also add:

```
echo "source ~/data/ros2_ws/install/setup.bash" >> ~/.bashrc
```

To apply changes without opening a new terminal:

```
source ~/.bashrc
```

19 Useful ROS 2 Workspace Commands

A typical workflow in this course will look like this.

Go to your workspace:

```
cd ~/data/ros2_ws
```

Check what is inside:

```
ls
```

Source ROS 2:

```
source /opt/ros/jazzy/setup.bash
```

Build your packages:

```
colcon build
```

Source the workspace:

```
source install/setup.bash
```

Check whether ROS 2 can find a package:

```
ros2 pkg list
```

For a specific package:

```
ros2 pkg prefix urdf_tutorial
```

20 Useful Keyboard Shortcuts

Shortcut	Function
Tab	Complete commands and paths
Up Arrow	Previous command
Down Arrow	Next command in history
Ctrl+C	Stop the running program
Ctrl+R	Search command history
Ctrl+L	Clear the terminal screen
Ctrl+Shift+C	Copy from the terminal
Ctrl+Shift+V	Paste into the terminal

Note that in many Ubuntu terminals, copying and pasting use Ctrl+Shift+C and Ctrl+Shift+V, rather than Ctrl+C and Ctrl+V. This is because Ctrl+C is used to stop a running terminal program.

21 Quick Reference

Command	Purpose
<code>pwd</code>	Show current directory
<code>ls</code>	List files and directories
<code>cd DIR</code>	Change directory
<code>cd ..</code>	Move up one directory
<code>cd ~</code>	Go to home directory
<code>mkdir DIR</code>	Create a directory
<code>mkdir -p PATH</code>	Create nested directories
<code>touch FILE</code>	Create an empty file
<code>cat FILE</code>	Display a file
<code>cp</code>	Copy files
<code>mv</code>	Move or rename files
<code>rm</code>	Delete files
<code>sudo apt install</code>	Install Ubuntu packages
<code>git clone</code>	Download a Git repository
<code>source</code>	Load environment settings
<code>colcon build</code>	Build a ROS 2 workspace

22 Practice Exercise

Before starting the ROS 2 assignments, make sure you can perform the following operations without using the graphical file manager.

1. Open a terminal.
2. Go to your home directory.
3. Create a directory named `data`.
4. Create `ros2_ws/src` inside `data`.
5. Navigate to the `src` directory.
6. Use `pwd` to verify your current directory.
7. Use `ls` to inspect the directory.
8. Clone the `urdf_tutorial` repository.
9. Return to the root of the ROS 2 workspace.
10. Source ROS 2 Jazzy.
11. Build the workspace using `colcon build`.
12. Source the newly built workspace.

If you are comfortable with these operations, you know the main Ubuntu terminal commands needed for this course.